$$\left[\frac{1}{T}\int_0^T dt\, e^{iHt} A e^{-iHt}, H \right] = 0$$

$\langle A, B \rangle = T_r(A^* B)$

$A = B + C$ when $[B, H] = 0$ and C doesn't

$A - [A,H] - [[A,H],H] - [[[A,H],H],H]$?

$\frac{1}{T}\int_0^T dt\, e^{iHt} |\psi\rangle$ = proj of $|\psi\rangle$ onto kernel of H

Reps of $\underline{SO(3)}$ $\quad$ <u>generators</u> $\quad X_x, X_y, X_z$

$[X_x, X_y] = X_z$

$[X_y, X_z] = X_x$

$[X_z, X_x] = X_y$

$e^{\theta X_z}$ rotation by θ along z axis

→ physicists want this to be $e^{-i\theta J_z}$ → if $J_z^\dagger = J_z$

$\left(e^{-i\theta J_z}\right)^\dagger = e^{i\theta J_z} = \left(e^{-i\theta J_z}\right)^{-1}$

- looking for finite-dim, irreducible reps, complex

we can complexify the Lie algebra

$\left(\underline{SO(3)}_{\mathbb{C}} \cong \underline{SO(2,1)}_{\mathbb{C}} \right)$

$J_x = i X_x \qquad J_y = i X_y \qquad J_z = i X_z$

$[J_x, J_y] = i J_z$

$[J_y, J_z] = i J_x$

$[J_z, J_x] = i J_y$

change of basis $\quad J_+ = J_x + i J_y$

$J_- = J_x - i J_y$

$[J_z, J_+] = J_+$

$[J_z, J_-] = -J_-$

$[J_+, J_-] = 2 J_z$

<u>Lemma</u> if $A : V \to V$ linear map $\dim V > 0$ then A has at least one non-zero eigenvector

Lemma if V fin. dimensional complex vector space, $\dim(V) > 0$.

if $A, B \in \mathrm{End}(V)$ with $[A, B] = zB$ $z \neq 0$

Then B is nilpotent and there is an eigenvector v of A s.t. $Bv = 0$ $(v \neq 0)$

$\exists\, k \in \mathbb{N}$ s.t $B^k = 0$

---

proof suppose $w \neq 0$ is an eigenvector of A, $Aw = aw$

$$AB\,w = (BA + AB - BA)\,w = BAw + [A,B]\,w = a\,Bw + z\,Bw = (a+z)\,Bw$$

so Bw is also an eigenvector of A with eigenvalue $a + z \neq a$

we know V is finite-dim so at some point $B^k w = 0$ otherwise we would have infinitely many eigenvectors with distinct eigenvalues.

if k smallest number s.t. $B^k w = 0$ then $B^{k-1} w \neq 0$

$B(B^{k-1} w) = 0$ so B has an eigenvector with 0-eigenvalue

$\dim \ker(B) \geq 1$

$$\begin{pmatrix} 0 & \cdots & 0 \\ \vdots & & \\ \vdots & & \\ \vdots & & \end{pmatrix}$$

Shur's lemma if V complex vector space with an irreducible action of a Lie algebra $\mathfrak{g}$ on it.

if $A: V \to V$ is a linear map that commutes with the action of the Lie algebra $([A, X] = 0 \;\; \forall X \in \mathfrak{g})$

$[A, \rho(X)]$

then A must be a multiple of identity.

$A(X \triangleright v) - X \triangleright (Av) = 0$

rep: $\rho : X \in \underline{g} \mapsto \rho(X) \in \mathrm{End}(V, V)$

think of $\rho(X) v$ as action of X on v

action of X on v, $X \triangleright v = \rho(X) v$

linear action $X \triangleright \alpha v + \beta w = \alpha X \triangleright v + \beta X \triangleright w$

---

proof: A has at least 1 eigenvalue, call it λ $(\exists v \neq 0, Av = \lambda v)$

$A' = A - \lambda \mathbb{1}$ $\dim(\ker(A')) \geq 1$

$A' v = 0$

if $X \in \underline{g}$ then $[\rho(X), A'] = [\rho(X), A] - [\rho(X), \lambda \mathbb{1}] = 0$

if $w \in \ker(A')$ then $A' \rho(X) w = \rho(X) A' w = 0$

so $\rho(X) w \in \ker(A')$

$\ker(A')$ invariant subspace

irreducible means that the only invariant subspaces are $\{0\}$ and V

- if V has dim 1 then A is necessarily a multiple of $\mathbb{1}$
- if $\dim(V) > 1$ then it must be $V = \ker(A')$

$A' v = 0 \quad \forall v \in V$ $(A - \lambda \mathbb{1}) v = 0 \quad \forall v \Rightarrow A = \lambda \mathbb{1}$

---

in our case $\dim V = n$ $n \geq 1$, action of J_+, J_-, J_z on V is irreducible

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$[J_z, J_+] = J_+ \Rightarrow$ there is some vector $|\psi\rangle \neq 0$ s.t. $J_+ |\psi\rangle = 0$

$J_z |\psi\rangle = j |\psi\rangle$

$j \in \mathbb{C}$

$[J_z, J_-] = -J_-$

$$J_z J_- |\psi\rangle = (J_- J_z + [J_z, J_-])|\psi\rangle$$
$$= J_- J_z |\psi\rangle - J_- |\psi\rangle$$
$$= (j-1) J_- |\psi\rangle$$

since V is finite-dim, there must be $k \in \mathbb{N}_0$ s.t $J_-^k |\psi\rangle = 0$

if k is the smallest such integer,

$$W = \text{space}\{ J_-^\ell |\psi\rangle \mid \ell = 0, 1, \dots, k-1 \} \qquad \dim(W) = k$$

<u>W is invariant</u> (made of eigenvectors of J_z so $J_z W = W$)

$$J_- W \subseteq W$$

$$J_+ W = ?$$

$$J_+ (J_-^k |\psi\rangle) = J_+ J_- (J_-^{k-1} |\psi\rangle)$$
$$= J_- J_+ (J_-^{k-1} |\psi\rangle) + [J_+, J_-](J_-^{k-1} |\psi\rangle)$$
$$= J_- J_+ (J_-^{k-1} |\psi\rangle) + 2 J_z J_-^{k-1} |\psi\rangle$$
$$= \dots J_-^2 J_+ J_-^{k-2} |\psi\rangle + \dots$$

W invariant, $W \neq \{0\} \Rightarrow W = V$

there is a basis of eigenvectors of J_z, each with distinct eigenvalue

basis of eigenvectors is $\{ |m\rangle \mid m = j - \ell, \ \ell = 0, \dots, n-1 \}$

$$J_z |m\rangle = m |m\rangle$$